Assignment 1: Probability & Random Variables

Part I: Theoretical Problems

Section A: Axioms, Set Theory, and Limits

1. Let $\mathbb{P}$ be a valid probability measure on a sample space Ω. If $\{A_n\}_{n=1}^{\infty}$ is a sequence of events such that $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$, use the axioms of probability (specifically countable additivity and Boole's inequality) to prove that the probability of infinitely many A_n occurring simultaneously is exactly 0.

2. Generalize Bonferroni's inequality to find the tightest possible lower bound for $\mathbb{P}(\bigcap_{i=1}^{n} A_i)$ given that $\mathbb{P}(A_i) = p_i$ for all i. Formulate a specific scenario where this lower bound is exactly achieved.

3. Let A and B be events in a sample space. Prove or disprove: If $\mathbb{P}(A|B) \geq \mathbb{P}(A)$, then it must be true that $\mathbb{P}(A^c|B^c) \geq \mathbb{P}(A^c)$.

Section B: Combinatorics & Classical Probability

4. **Generalized Capture-Recapture:** In a wildlife reserve, an unknown population of N animals exists. Initially, T animals are captured, tagged, and released. Later, a random sample of n animals is captured without replacement, and exactly k of them are found to be tagged. Show analytically that the value of N that maximizes the probability of this specific outcome is $\lfloor \frac{nT}{k} \rfloor$.

5. **Asymptotic Derangements:** A professor drops a stack of n uniquely graded assignments and hands them back to the n students completely at random. Derive the exact probability that exactly k students receive their correct assignment. Then, rigorously prove that as $n \to \infty$, this probability converges to $\frac{1}{e \cdot k!}$.

6. **The Modified Polya Urn:** An urn initially contains r red and b blue balls. A ball is drawn at random, its color is noted, and it is replaced along with $c > 0$ balls of the *opposite* color. Let R_n be the event that the n-th draw is red. Formulate the exact probability of R_2 and R_3. Does $\mathbb{P}(R_n)$ remain time-invariant as it does in the standard Polya Urn model? Prove your claim.

Section C: Conditional Probability & Bayes

7. **The Three Prisoners Paradox:** Prisoners A, B, and C are on death row. The governor randomly pardons one. The warden knows who is pardoned but is not allowed to tell A his own fate. A begs the warden: "Name one of the other two who will be executed." The warden truthfully says "B will be executed." Prisoner A calculates his survival probability is now 1/2. Use Bayes' Theorem to rigorously prove whether A is correct or incorrect.

8. Two players, Alice and Bob, alternately toss a biased coin with probability p of Heads. Alice tosses first. The first to toss Heads wins the game. Let A be the event that Alice wins. Calculate $\mathbb{P}(A)$ strictly using mutually exclusive event expansions. Does there exist any bias $p \in (0,1)$ such that the game is perfectly fair?

9. Suppose A, B, and C are events such that $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C)$. Does this imply that A, B, and C are mutually independent? If so, prove it. If not, rigorously construct a sample space and probability measure that serves as a counter-example.

Section D: CDF Properties and Discontinuities

10. Let $F_X(x)$ be the CDF of a random variable X. Using the definition of limits of sets and the probability axioms, rigorously prove that the set of all discontinuity points of any valid CDF must be at most countable.

11. Construct a valid CDF $F(x)$ such that the random variable X is discrete, its support is the set of all rational numbers $\mathbb{Q}$ in $[0,1]$, and $F(x)$ is strictly increasing on $(0,1)$. (*Hint: Use an enumeration of the rationals*).

12. Let X be a random variable representing the straight-line distance from the origin to a point chosen uniformly at random within the 2D unit disk $x^2 + y^2 \leq 1$. Derive the CDF and PDF of X.

13. Let X be a random variable distributed uniformly on the interval $(-1,1)$, meaning its PDF is $f_X(x) = 1/2$ for $-1 < x < 1$. Let $Y = X^2$. Find the CDF and PDF of Y.

Section E: PDFs, Quantiles, and Mixed Random Variables

14. Prove or disprove: "A valid Probability Density Function $f(x)$ for a continuous random variable must be bounded above." If true, prove it. If false, construct a valid PDF $f(x)$ that is unbounded at infinitely many points within the interval $[0,1]$.

15. Let X_1 and X_2 be two independent random variables drawn from the same continuous CDF $F(x)$. Prove, using the definition of continuity and sequence limits, that $\mathbb{P}(X_1 = X_2) = 0$.

16. Let X have a mixed distribution where with probability $p \in (0,1)$, $X = 0$, and with probability $1 - p$, X follows a continuous distribution with a strictly increasing CDF $G(x)$ for $x > 0$. Express the general formula for the q-th quantile, ξ_q, of X in terms of p and $G(x)$.

Part II: Computational & Simulation Problems

Instructions: For this section, you can implement the simulations using Python, C++, or any language you feel comfortable in. Your submission must include the commented code, resulting visualizations, and a mathematical discussion of the empirical results.

Simulation 1: Validating the Capture-Recapture Distribution

Assume a true animal population of $N = 1000$. You capture and tag $T = 100$ animals. Later, you capture a sample of $n = 150$ animals.

1. Write a script to simulate this second capture process 10,000 times without replacement.
2. Track k, the number of tagged animals found in each simulation.
3. Plot a normalized histogram of your simulated values of k (the empirical PMF).
4. Compute the theoretical PMF using classical probability (combinations) and overlay it on your plot. Discuss the match.

Simulation 2: Penney's Game (Non-Transitive Probabilities)

Let X be the event that the sequence `HTT` appears before `HTH` in a series of independent fair coin tosses.

1. Write a simulation that flips coins until either `HTT` or `HTH` appears. Run this 100,000 times to find the empirical probability that `HTT` wins.
2. Now, pit `HTH` against `THH`, and `THH` against `TTH`. Find their empirical win rates.
3. Discuss the paradox: Since every specific 3-toss window has exactly a $1/8$ chance of appearing, explain purely via sequence overlaps and conditional probabilities why these head-to-head races are not 50/50, and why the "winning" relation is non-transitive.